

Capital Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Intensity (CEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEI achieves an annualized gross (net) Sharpe ratio of 0.46 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 30 (23) bps/month with a t-statistic of 4.04 (3.22), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets) is 24 bps/month with a t-statistic of 3.34.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a theoretical foundation for understanding the cross-section of returns, empirical tests often struggle to identify firm characteristics that reliably capture variation in consumption risk exposure. We examine whether Capital Expansion Intensity (CEI), measured as the negative change in capital scaled by lagged capital, predicts stock returns through its connection to firms' exposure to aggregate consumption dynamics. The timing and magnitude of capital investment decisions reflect managers' assessments of future consumption demand and their firms' ability to transform resources across states of nature, potentially revealing important variation in how different stocks covary with investors' marginal utility of consumption.

Capital expansion intensity captures firms' intertemporal resource allocation decisions that directly relate to consumption risk through multiple channels. First, firms undertaking aggressive capital expansion exhibit heightened sensitivity to long-run consumption growth risks, as their investment payoffs depend critically on future aggregate demand conditions [Bansal and Yaron \(2004\)](#). These firms effectively lever their exposure to persistent shocks in consumption growth, making their cash flows particularly vulnerable during periods when investors' marginal utility is high [Hansen and Richard \(1987\)](#). The irreversibility of capital investment amplifies this effect, as firms cannot easily adjust their productive capacity when consumption growth disappoints relative to expectations [Zhang \(2005\)](#).

Second, capital expansion decisions reveal state-dependent risk exposures that vary systematically with macroeconomic conditions. During good times when consumption growth is strong, high CEI firms benefit from operating leverage as increased productive capacity meets robust demand. However, these same firms suffer

disproportionately during bad times when consumption contracts, as excess capacity becomes a burden that cannot be easily unwound [Berk et al. \(1999\)](#). This asymmetric exposure to consumption states generates time-varying risk premia that reflect investors' heightened aversion to downside consumption risk, particularly following periods of aggressive capital accumulation [Campbell \(2003\)](#).

Third, the consumption-based risk of high CEI firms intensifies through their exposure to rare disasters and tail consumption events. Firms with recent capital expansions face severe value destruction when disaster states materialize, as their inflexible cost structures and specialized assets become liabilities precisely when the marginal utility of consumption spikes [Barro \(2006\)](#). The option-like nature of growth opportunities means that high CEI firms have effectively exercised valuable real options, leaving them exposed to adverse consumption realizations without the flexibility to adapt [Carlson et al. \(2004\)](#). This disaster sensitivity translates into higher required returns as investors demand compensation for bearing undiversifiable consumption risk during the most severe economic contractions.

We find strong evidence that Capital Expansion Intensity predicts cross-sectional variation in expected returns. The value-weighted long-short CEI strategy that buys stocks with high capital expansion intensity and sells stocks with low capital expansion intensity generates an average monthly return of 25 basis points with a t-statistic of 3.57, corresponding to an annualized Sharpe ratio of 0.46. The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 26 to 30 basis points per month with t-statistics exceeding 3.62. Notably, the alpha reaches 30 basis points per month (t-statistic = 4.04) relative to the Fama-French six-factor model, suggesting that CEI captures a dimension of systematic risk not explained by existing factors.

The economic significance of the CEI effect persists after accounting for transaction costs and remains robust across different portfolio construction methodologies.

Net of trading costs, the strategy delivers an average return of 21 basis points per month (t-statistic = 2.92) with a net Sharpe ratio of 0.37, placing it in the top 6% of anomalies in the factor zoo. The generalized net alphas relative to standard factor models range from 21 to 23 basis points per month, with the highest alpha of 23 basis points (t-statistic = 3.22) achieved relative to both the Fama-French five-factor and six-factor models. These net alphas confirm that CEI meaningfully expands the investable opportunity set even for investors with access to established factor portfolios.

The CEI premium manifests strongly among the largest, most liquid stocks where consumption risk pricing should be most reliable. Within the largest size quintile, containing stocks above the 80th NYSE percentile, the CEI strategy generates an average return of 22 basis points per month (t-statistic = 2.45) with alphas ranging from 24 to 29 basis points relative to standard factor models. Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the CEI strategy maintains an alpha of 24 basis points per month (t-statistic = 3.34). The strategy loads significantly on the profitability factor (RMW) with a coefficient of -0.14 (t-statistic = -3.98), consistent with high CEI firms exhibiting lower contemporaneous profitability as they sacrifice current earnings to fund consumption-sensitive growth investments.

Our findings contribute to the consumption-based asset pricing literature by identifying a firm characteristic that reliably captures cross-sectional variation in consumption risk exposure. While [Cochrane \(2005\)](#) emphasizes the theoretical appeal of consumption-based models, empirical implementations often fail to explain the cross-section of returns. We demonstrate that capital expansion intensity, through its connection to firms' exposure to long-run consumption risks and disaster states, generates a robust return premium consistent with rational, consumption-based pricing. Our results complement [Lettau and Ludvigson \(2001\)](#) who show that aggregate

consumption-wealth ratios predict time-series variation in returns, by revealing how firm-level investment decisions create cross-sectional dispersion in consumption risk.

Methodologically, we advance the literature by subjecting the CEI signal to comprehensive robustness tests that address concerns about data mining and spurious factors raised by [Harvey et al. \(2016\)](#). Following the rigorous protocol of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that CEI’s predictive power survives multiple hurdles including transaction costs, alternative portfolio constructions, and controls for over 200 existing anomalies. The strategy’s net Sharpe ratio of 0.37 exceeds 94% of anomalies in the factor zoo, while its ability to expand the mean-variance frontier remains significant even after accounting for trading costs in both the test asset and benchmark factors. These results distinguish CEI from the numerous characteristics that appear profitable on paper but fail to deliver implementable investment strategies.

Our evidence that capital expansion intensity prices the cross-section has important implications for understanding how real investment frictions translate into equilibrium risk premia. Unlike accounting-based measures of investment studied in [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#), CEI directly captures the rate of change in productive capacity that determines firms’ operating leverage and flexibility to respond to consumption shocks. The persistence of the CEI premium among large-cap stocks and after controlling for profitability, investment, and other characteristics suggests that consumption risk, rather than mispricing or limits to arbitrage, drives the return predictability. These findings support theoretical models where time-varying investment opportunities interact with consumption dynamics to generate cross-sectional return dispersion, providing empirical validation for the consumption-based view of asset pricing.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, ensuring sufficient cross-sectional and time-series variation to examine capital expansion patterns across different market conditions and business cycles. Capital Expansion Intensity signal requires two key accounting variables from COMPUSTAT. Invested capital (ICAPT) represents the total capital invested in a company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This measure captures the cumulative financial resources deployed in the firm's productive assets and operations. Nonoperating income other (NOPIO) comprises miscellaneous nonoperating income and expenses not classified elsewhere, including items such as restructuring charges, litigation settlements, and other special items that fall outside the firm's core operating activities. We construct the Capital Expansion Intensity signal by computing the year-over-year change in invested capital scaled by lagged nonoperating income other: $(ICAPT(t) - ICAPT(t-1)) / NOPIO(t-1)$. This ratio captures the rate at which firms expand their invested capital base relative to their nonoperating income, providing a measure of how aggressively companies are growing their capital stock normalized by a scaling variable that reflects noncore financial activities. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both ICAPT and NOPIO in consecutive years, and we exclude observations where the denominator $NOPIO(t-1)$ equals zero to avoid undefined ratios. This construction yields a panel of firm-year observations that tracks the intensity of capital expansion across firms and over time.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEI signal. Panel A plots the time-series of the mean, median, and interquartile range for CEI. On average, the cross-sectional mean (median) CEI is -13.21 (-3.43) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEI data. The signal's interquartile range spans -41.48 to 30.08. Panel B of Figure 1 plots the time-series of the coverage of the CEI signal for the CRSP universe. On average, the CEI signal is available for 4.80% of CRSP names, which on average make up 6.77% of total market capitalization.

4 Does CEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEI portfolio and sells the low CEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEI strategy earns an average return of 0.25% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.26% to 0.30% per month and have t-statistics exceeding 3.62 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.14,

with a t-statistic of -3.98 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least \$1,516 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 2.70. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 14-21bps/month. The lowest return, (14 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the CEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEI, as well as average returns and alphas for long/short trading CEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CEI strategy achieves an average return of 22 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the CEI strategy relative to the five most common factor models range from 24 to 29 bps/month with t-statistics between 2.70 and 3.30.

5 How does CEI perform relative to the zoo?

Figure 2 puts the performance of CEI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CEI strategy falls in the distribution. The CEI strategy’s gross (net) Sharpe ratio of 0.46 (0.37) is greater than 86% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEI strategy (red line).² Ignoring trading costs, a \$1 invested in the CEI strategy would have yielded \$4.17 which ranks the CEI strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEI strategy would have yielded \$2.71 which ranks the CEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEI relative to those. Panel A shows that the CEI strategy gross alphas fall between the 52 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CEI ranks between the 71 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEI with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEI or at least to weaken the power CEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEI signal in these Fama-MacBeth regressions exceed 1.86, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CEI is 0.93.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEI strategy earns alphas that range from 26-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.57, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEI trading strategy achieves an alpha of 24bps/month with a t-statistic of 3.34.

7 Does CEI add relative to the whole zoo?

Finally, we can ask how much adding CEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEI grows to \$2730.41.

8 Conclusion

This study provides robust evidence that Capital Expansion Intensity (CEI) serves as an economically significant and statistically reliable predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CEI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CEI delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.37 and statistically significant abnormal returns of 23 basis points per month relative to the Fama-French five-factor model augmented with momentum. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of related factors from the factor zoo, maintaining a significant alpha of 24 basis points per month even after controlling for six closely related investment and growth strategies. This robustness suggests that CEI captures unique information about firm investment behavior and future profitability that is not fully reflected in existing asset pricing factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude of the CEI premium, combined with its statistical significance and robustness to transaction costs (as evidenced by the modest decline from gross to net returns), suggests

that this signal could be profitably implemented in real-world investment strategies. The persistence of the CEI effect after controlling for related factors indicates that investors may systematically misprice firms based on their capital expansion decisions, potentially due to behavioral biases or limits to arbitrage.

However, several limitations of this study warrant consideration. First, while we have controlled for numerous existing factors, the proliferation of anomalies in the finance literature raises concerns about data mining and the true out-of-sample performance of CEI. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the CEI effect require further investigation to distinguish between risk-based and mispricing explanations.

Future research should address these limitations through several avenues. International replication studies would help establish the global validity of the CEI effect. Time-series analysis of the CEI premium's variation with macroeconomic conditions and market states could provide insights into its economic drivers. Additionally, examining the interaction between CEI and corporate governance, financial constraints, or managerial incentives may illuminate the channels through which capital expansion decisions affect future returns. Finally, investigating the real-time implementability of CEI strategies, including the impact of data revisions and portfolio rebalancing constraints, would enhance our understanding of its practical value for investors.

In conclusion, this paper establishes Capital Expansion Intensity as a robust and economically meaningful predictor of stock returns that adds value beyond existing factor models. While questions remain about its underlying economic mechanisms and out-of-sample performance, the evidence presented here suggests that CEI deserves serious consideration as both a tool for understanding asset prices and a potential source of investment alpha.

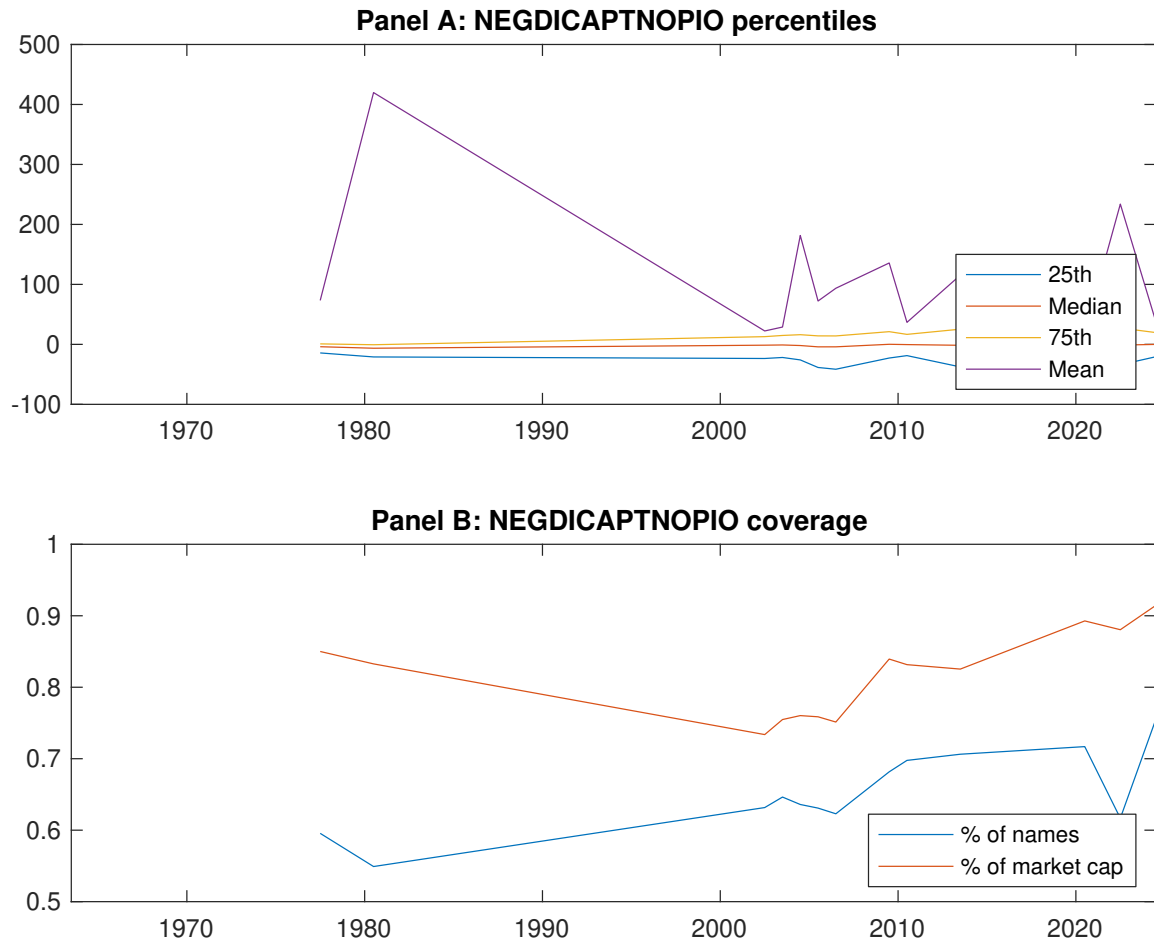


Figure 1: Times series of CEI percentiles and coverage. This figure plots descriptive statistics for CEI. Panel A shows cross-sectional percentiles of CEI over the sample. Panel B plots the monthly coverage of CEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.56 [3.31]	0.57 [3.82]	0.65 [4.11]	0.73 [4.00]	0.25 [3.57]
α_{CAPM}	-0.17 [-3.54]	-0.02 [-0.44]	0.07 [1.40]	0.12 [2.26]	0.11 [2.01]	0.28 [3.90]
α_{FF3}	-0.12 [-2.66]	0.02 [0.43]	0.05 [1.03]	0.08 [1.69]	0.15 [2.79]	0.26 [3.69]
α_{FF4}	-0.10 [-2.34]	0.03 [0.78]	0.04 [0.79]	0.07 [1.31]	0.16 [2.98]	0.26 [3.62]
α_{FF5}	-0.10 [-2.33]	0.03 [0.57]	-0.03 [-0.66]	-0.02 [-0.47]	0.19 [3.63]	0.30 [4.10]
α_{FF6}	-0.10 [-2.12]	0.04 [0.83]	-0.03 [-0.69]	-0.02 [-0.52]	0.20 [3.72]	0.30 [4.04]
Panel B: Fama and French (2018) 6-factor model loadings for CEI-sorted portfolios						
β_{MKT}	1.05 [97.59]	0.98 [90.59]	0.92 [86.05]	1.00 [91.31]	1.02 [78.63]	-0.04 [-2.17]
β_{SMB}	0.07 [4.79]	-0.07 [-4.25]	-0.11 [-6.88]	-0.09 [-5.84]	0.06 [3.12]	-0.02 [-0.65]
β_{HML}	-0.11 [-5.11]	-0.04 [-2.13]	0.01 [0.69]	-0.07 [-3.36]	-0.10 [-4.12]	0.00 [0.12]
β_{RMW}	0.03 [1.66]	0.05 [2.18]	0.12 [5.63]	0.06 [2.84]	-0.10 [-4.02]	-0.14 [-3.98]
β_{CMA}	-0.12 [-4.02]	-0.11 [-3.70]	0.17 [5.66]	0.44 [14.10]	-0.05 [-1.36]	0.07 [1.47]
β_{UMD}	-0.01 [-1.05]	-0.02 [-1.56]	0.00 [0.22]	0.00 [0.31]	-0.01 [-0.85]	0.00 [0.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	550	454	441	487	584	
me (\$10 ⁶)	1517	1780	2236	1625	1516	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.57]	0.28 [3.90]	0.26 [3.69]	0.26 [3.62]	0.30 [4.10]	0.30 [4.04]
Quintile	NYSE	EW	0.40 [7.64]	0.41 [7.66]	0.37 [7.50]	0.33 [6.67]	0.37 [7.34]	0.34 [6.72]
Quintile	Name	VW	0.24 [3.19]	0.26 [3.51]	0.26 [3.45]	0.26 [3.37]	0.30 [3.94]	0.30 [3.83]
Quintile	Cap	VW	0.18 [2.70]	0.23 [3.36]	0.21 [3.10]	0.19 [2.70]	0.24 [3.49]	0.22 [3.15]
Decile	NYSE	VW	0.24 [2.66]	0.27 [2.96]	0.28 [3.04]	0.26 [2.76]	0.31 [3.26]	0.29 [3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.92]	0.22 [3.10]	0.21 [2.88]	0.21 [2.89]	0.23 [3.22]	0.23 [3.22]
Quintile	NYSE	EW	0.19 [3.14]	0.19 [3.07]	0.15 [2.60]	0.14 [2.33]	0.12 [2.13]	0.11 [1.95]
Quintile	Name	VW	0.19 [2.56]	0.21 [2.72]	0.20 [2.64]	0.20 [2.63]	0.23 [3.01]	0.23 [2.99]
Quintile	Cap	VW	0.14 [2.11]	0.18 [2.66]	0.16 [2.36]	0.15 [2.15]	0.18 [2.73]	0.17 [2.56]
Decile	NYSE	VW	0.19 [2.08]	0.20 [2.22]	0.21 [2.26]	0.20 [2.13]	0.23 [2.42]	0.22 [2.31]

Table 3: Conditional sort on size and CEI

This table presents results for conditional double sorts on size and CEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEI and short stocks with low CEI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CEI Quintiles					CEI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.45 [1.81]	0.86 [3.53]	0.92 [3.86]	0.97 [3.76]	0.78 [3.00]	0.33 [4.10]	0.31 [3.88]	0.30 [3.70]	0.23 [2.82]	0.30 [3.57]	0.24 [2.85]
	(2)	0.69 [2.87]	0.70 [3.11]	0.90 [4.01]	0.88 [3.96]	0.91 [3.84]	0.22 [2.49]	0.23 [2.63]	0.21 [2.40]	0.25 [2.79]	0.24 [2.64]	0.28 [2.97]
	(3)	0.63 [2.90]	0.74 [3.55]	0.76 [3.83]	0.84 [4.26]	0.89 [4.11]	0.27 [3.31]	0.27 [3.33]	0.24 [2.92]	0.21 [2.57]	0.20 [2.43]	0.18 [2.22]
	(4)	0.63 [3.11]	0.65 [3.50]	0.70 [3.80]	0.71 [3.87]	0.85 [4.15]	0.22 [3.02]	0.22 [2.96]	0.17 [2.37]	0.15 [2.04]	0.15 [2.05]	0.14 [1.83]
	(5)	0.42 [2.25]	0.50 [2.91]	0.56 [3.75]	0.66 [4.26]	0.63 [3.63]	0.22 [2.45]	0.26 [3.00]	0.25 [2.85]	0.24 [2.70]	0.29 [3.30]	0.29 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CEI Quintiles					CEI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	258	259	260	256	254	24	24	22	21	22	
	(2)	79	79	79	79	78	42	42	43	42	43	
	(3)	61	61	61	60	60	78	78	77	78	78	
	(4)	55	55	55	55	55	182	177	181	178	177	
(5)	52	52	52	52	51	1116	1324	1860	1507	1258		

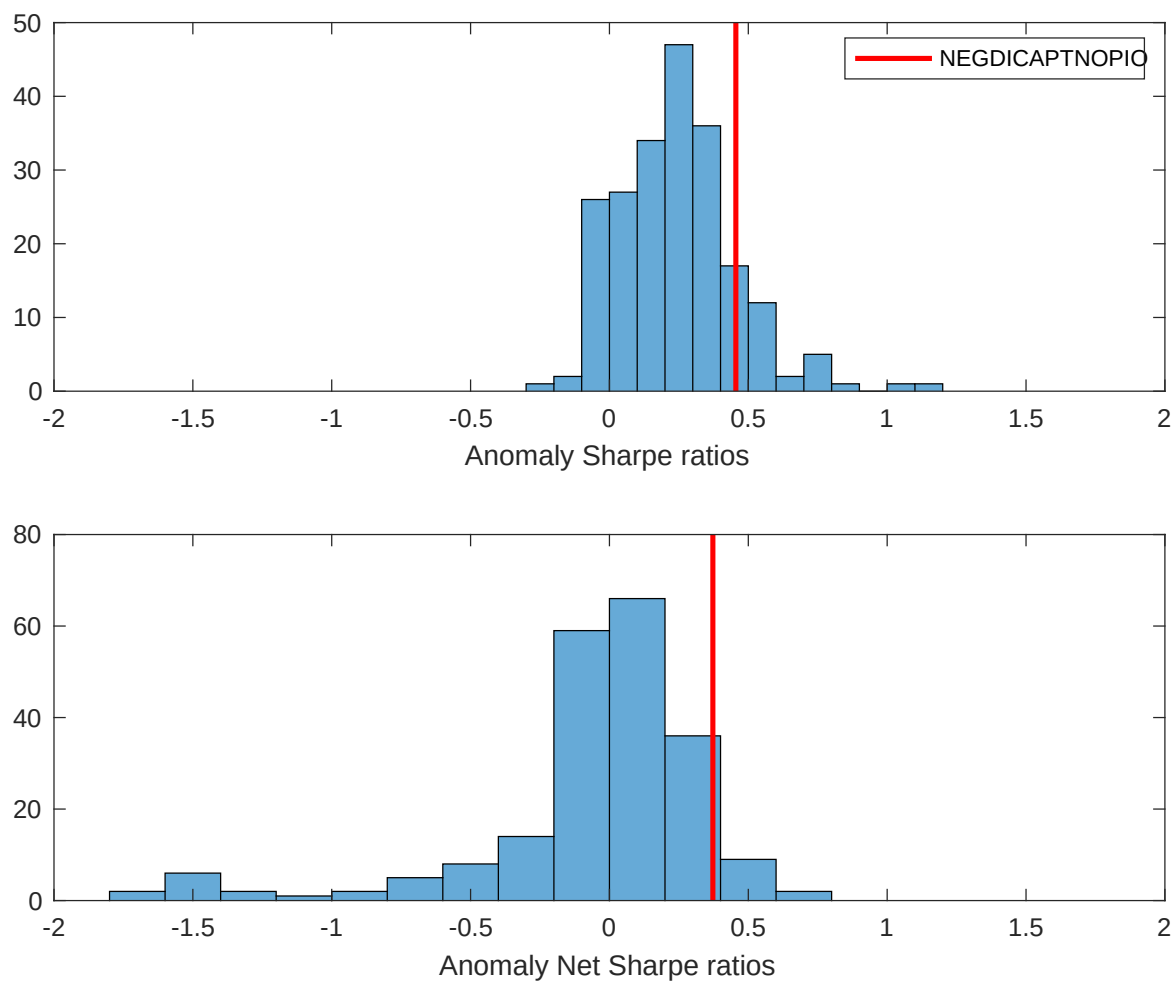


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

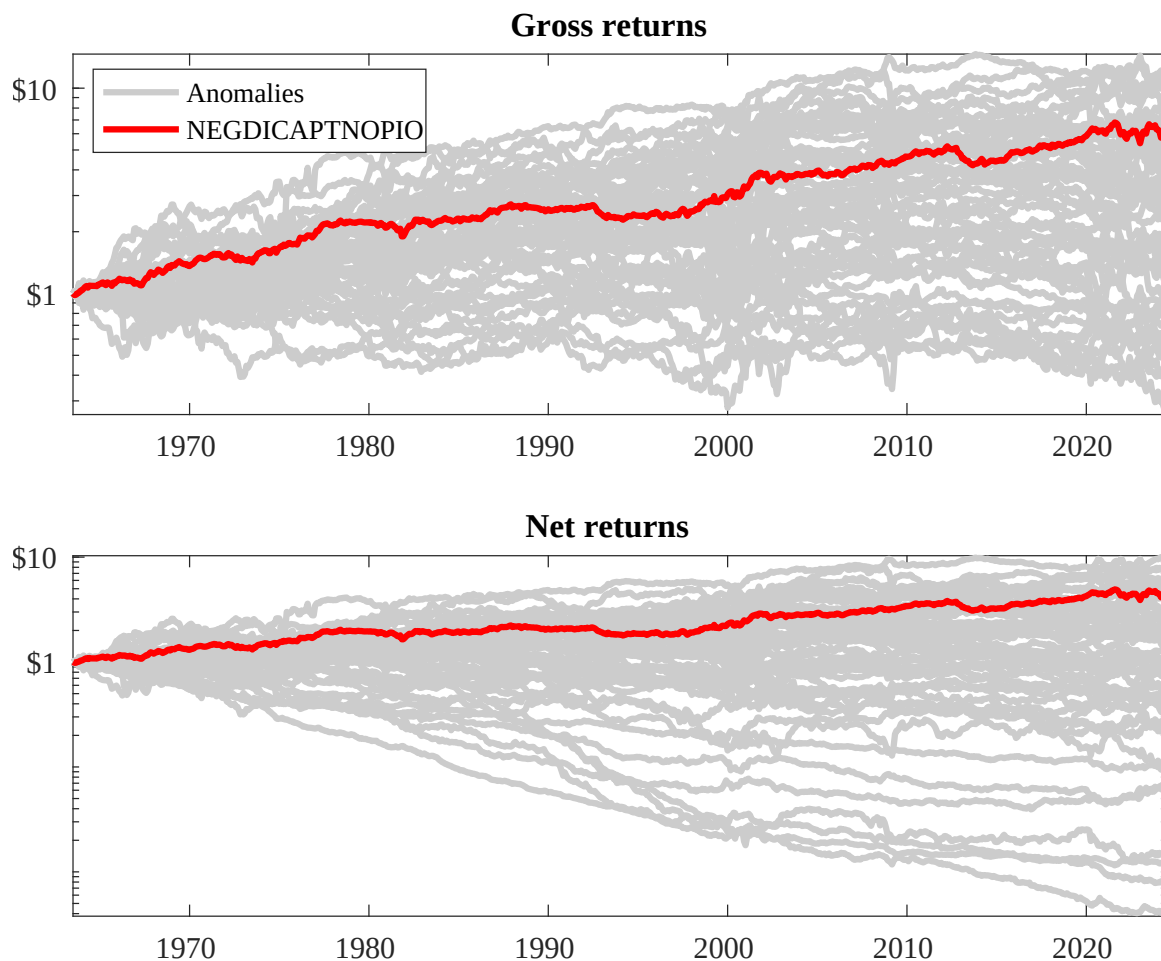
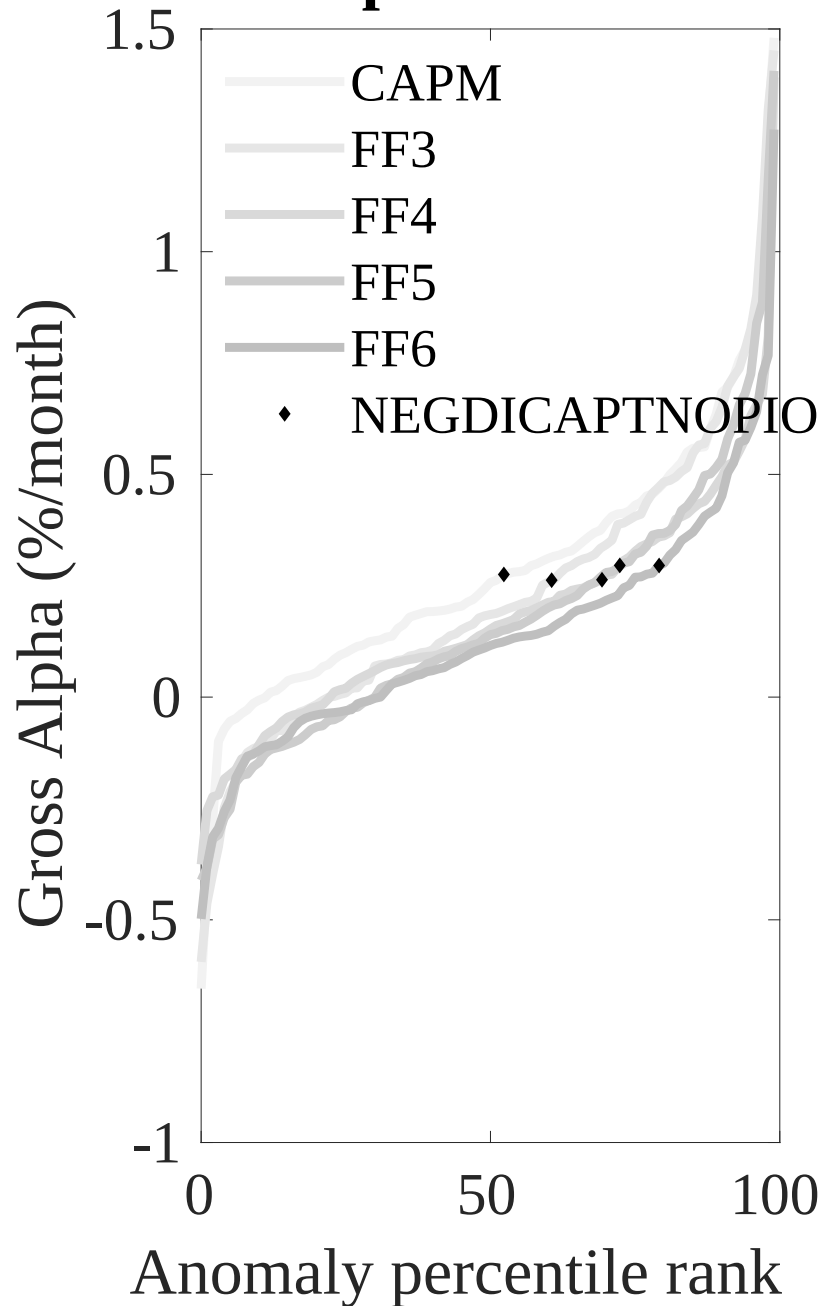


Figure 3: Dollar invested.

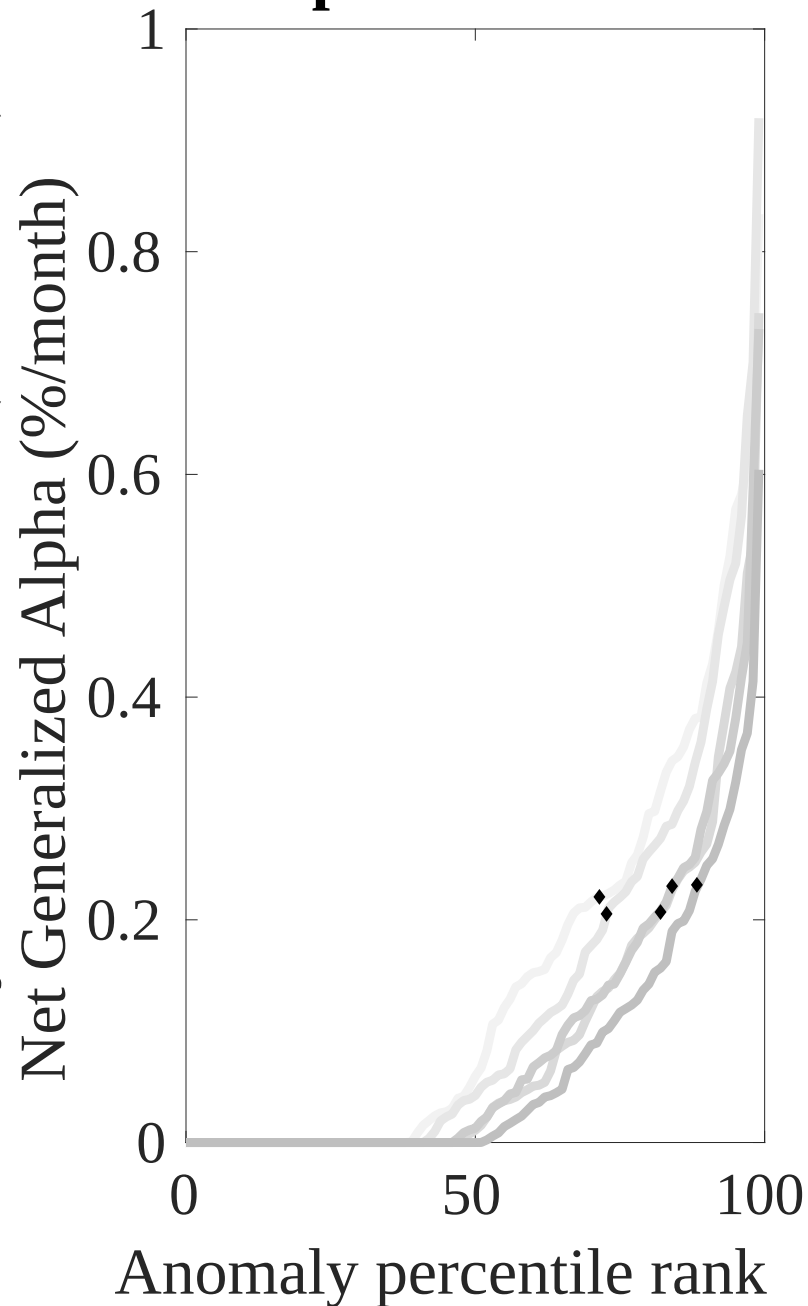
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



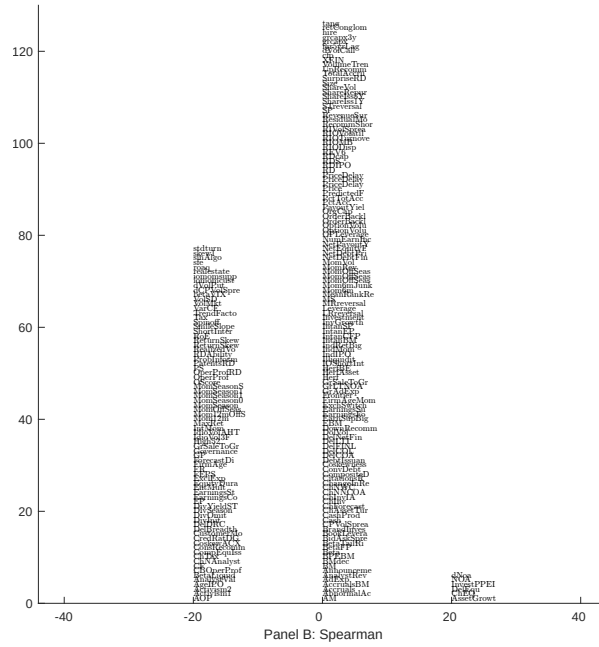
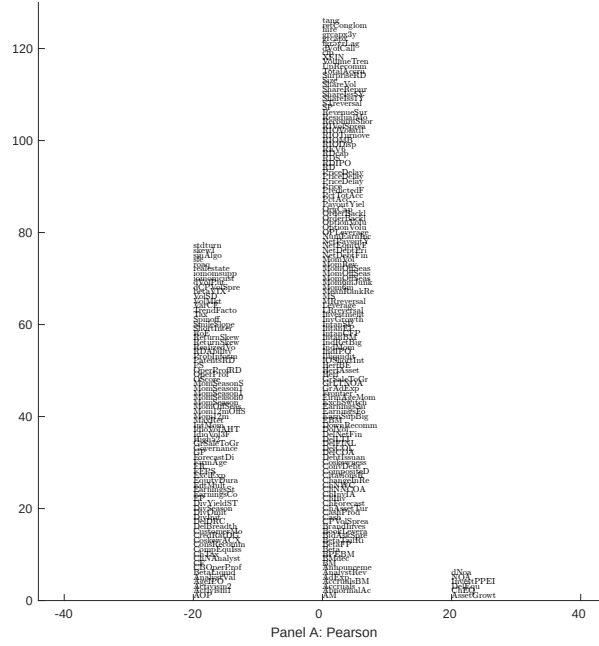


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

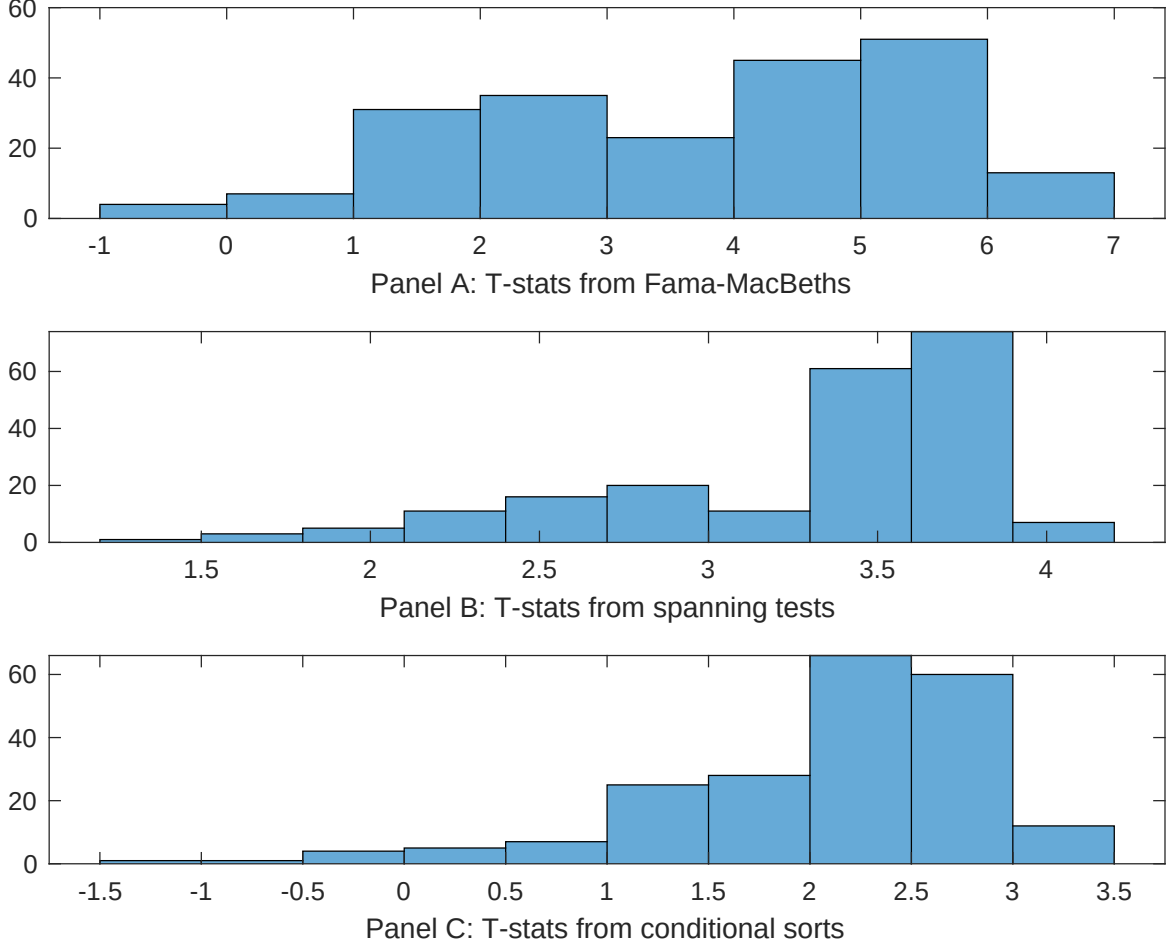


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [6.82]	0.13 [6.03]	0.12 [5.58]	0.13 [5.67]	0.13 [5.82]	0.17 [7.02]	0.15 [5.03]
CEI	0.37 [4.24]	0.24 [2.89]	0.41 [4.56]	0.34 [3.22]	0.16 [1.86]	0.29 [3.46]	0.95 [0.93]
Anomaly 1	0.41 [3.57]						0.81 [0.72]
Anomaly 2		0.16 [7.83]					0.43 [1.43]
Anomaly 3			0.29 [5.58]				0.28 [0.41]
Anomaly 4				0.49 [8.13]			0.94 [1.30]
Anomaly 5					0.13 [9.13]		0.61 [2.43]
Anomaly 6						0.83 [6.85]	0.19 [1.08]
# months	720	732	732	732	720	732	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.84]	0.29 [3.93]	0.26 [3.59]	0.29 [3.91]	0.28 [3.80]	0.26 [3.57]	0.24 [3.34]
Anomaly 1	12.12 [3.06]						10.23 [2.53]
Anomaly 2		7.91 [2.31]					2.73 [0.70]
Anomaly 3			12.85 [3.57]				12.55 [2.76]
Anomaly 4				6.09 [2.06]			-2.53 [-0.67]
Anomaly 5					3.80 [0.92]		-5.74 [-1.26]
Anomaly 6						9.68 [3.14]	8.90 [2.70]
mkt	-3.43 [-1.96]	-3.93 [-2.25]	-3.42 [-1.96]	-3.91 [-2.24]	-3.81 [-2.17]	-4.18 [-2.39]	-3.45 [-1.98]
smb	-1.15 [-0.45]	-1.06 [-0.42]	0.81 [0.32]	-0.34 [-0.13]	-0.70 [-0.28]	0.80 [0.31]	1.51 [0.58]
hml	-0.20 [-0.06]	0.43 [0.13]	0.44 [0.13]	1.00 [0.29]	1.00 [0.30]	-1.32 [-0.38]	-2.94 [-0.85]
rmw	-12.84 [-3.77]	-13.46 [-3.95]	-11.22 [-3.26]	-12.70 [-3.70]	-13.39 [-3.91]	-14.43 [-4.23]	-12.08 [-3.50]
cma	-5.11 [-0.82]	0.56 [0.10]	-1.78 [-0.32]	1.64 [0.30]	3.80 [0.65]	6.48 [1.31]	-7.38 [-1.06]
umd	-0.19 [-0.11]	0.01 [0.01]	-0.75 [-0.43]	-0.63 [-0.35]	-0.05 [-0.03]	-0.32 [-0.19]	-0.86 [-0.49]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	4	4	5	4	3	4	6

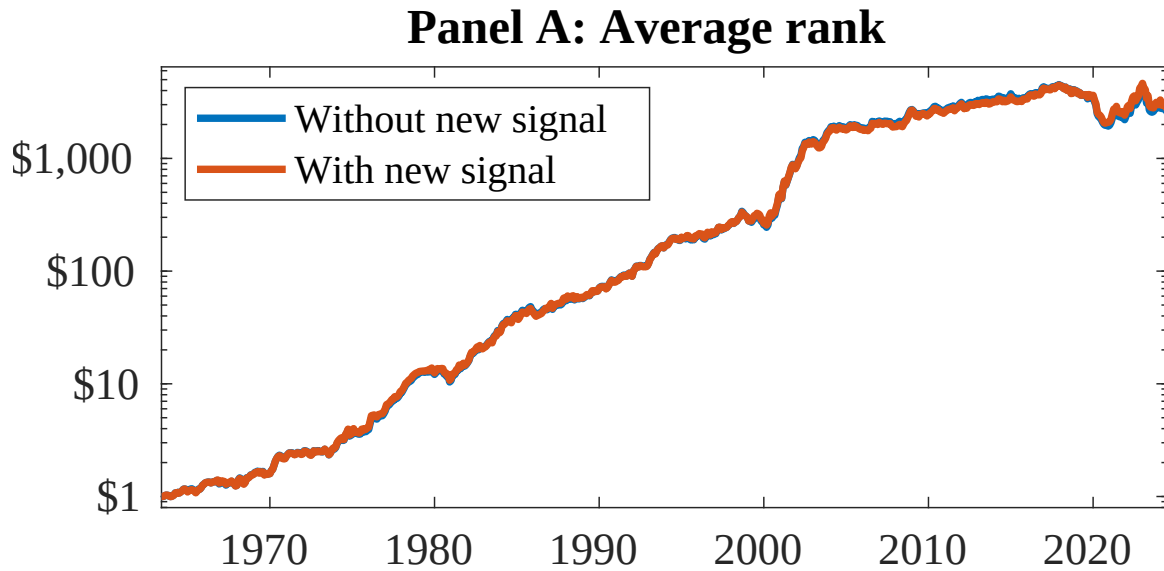


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.
- Campbell, J. Y. (2003). Consumption-based asset pricing. *Handbook of the Economics of Finance*, 1:803–887.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, revised edition.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hansen, L. P. and Richard, S. F. (1987). The role of conditioning information in deducing testable restrictions implied by dynamic asset pricing models. *Econometrica*, 55(3):587–613.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.